Geoglyph Dataset Pipeline — Process Log

A running record of decisions, methods, and results for the dataset-building phase of the thesis. Last updated: 2026-09-28.

1. Background: why the old dataset was abandoned

The project inherited a binary image dataset from the prior team (Zach and Colin): 530 "geoglyph" (positive) images and 2,181 "non-geoglyph" (negative) images, built from an unspecified combination of a KML file and an Excel catalog via Google Earth imagery.

Two problems surfaced during initial review:

- **Provenance was unclear.** It wasn't documented which version/date of the geoglyph catalog was used, or how the 530 positives were selected from a much larger pool of candidate coordinates.
- **Mislabeling was found by inspection.** Manual spot-checking of the "non-geoglyph" folder revealed real geoglyphs sitting in the negative class.

Steve (external collaborator) confirmed he used a catalog version dated 07/29/2025, working primarily in the Amazonas region for his own thesis, and provided:

- `amazon_geoglyphs.xlsx` — the canonical coordinate catalog (source: James Q. Jacobs' public Amazon geoglyphs archive)
- `amazon_geoglyphs.kml` — a KML export from Zach and Colin's model (later found to include Bolivia outliers and not match Steve's Amazonas-only subset)

Decision: manually correcting the mislabeled 530/2,181 image set was judged infeasible given the volume of images. Rather than partially fix a compromised dataset, we chose to rebuild the positive image set from scratch using the full xlsx coordinate catalog (1,477 raw rows) as the single source of truth going forward.

2. Coordinate data cleaning (Colab Notebook 1)

Starting from `amazon_geoglyphs.xlsx` (1,477 rows: idx, code, place, lat, lon, elev, source, a_width, b_width, form), a deep data-understanding and cleaning pass was run:

- **Valid coordinates:** 1,455 of 1,477 rows have non-null lat/lon.
- **Bounding-box sanity filter** (-25 to 5 lat, -75 to -55 lon) removed a known bad row (`acre34`, lat = lon = -67.808143 — a copy-paste-style data entry error) plus other out-of-region points, leaving 1,453 rows.

- **Missing-data audit:** `b_width` is 48.4% missing, but this is **structural, not a data gap** — shapes with a single defining dimension (circle, square, pentagon) simply don't have a second width.
- **Nearest-neighbor analysis** (BallTree, haversine distance) surfaced:
  - 2 exact duplicate `code` values (data entry duplicates) — dropped.
  - 8 near-duplicate coordinate pairs (<5m apart, different codes) — almost certainly the same physical site logged/reclassified twice. Each pair was resolved by keeping one site and logging the dropped one.
- **Regional grouping:** a 3-letter code-prefix (e.g. `acr`, `ama`, `ron`) maps informally to Brazilian states (Acre, Amazonas, Rondônia) and is retained as a grouping variable for future stratified/grouped cross-validation (`StratifiedGroupKFold`), to prevent same-site leakage across folds.

  Output: `amazon_geoglyphs_cleaned.csv` — **1,443 sites**, plus `duplicate_resolution_log.csv` documenting every dropped near-duplicate and why.

3. Imagery source and currency decision

Two imagery options were considered for downloading positive images:

- Official Google Maps Platform Static API (licensed, citable, but requires an API key/billing)
- Unofficial `mt0.google.com/vt` tile-stitching endpoint (no key, no license — used by the prior team's `v2gmapDownloader.py`)

Decision (interim): continue with the unofficial endpoint for now, since it requires no setup and worked reliably in testing. **Flagged as an open provenance concern to revisit before publication** — a reviewer may reasonably ask about the imagery license/citation.

A separate, important finding surfaced during testing (see Section 5): the catalog's `source` column (e.g. "GE 2021-11-20") indicates specific historical Google Earth imagery dates when each site was known to be visible — likely during a temporary forest-clearing event. The tile endpoint used here has **no date control** and serves whatever imagery is currently cached, which does not guarantee the same clearing is still visible today.

4. Positive-image downloader: design

Built `download_positives.py`, rewriting the prior team's `v2gmapDownloader.py` to fix several issues:

Issue in original script	Fix
Fixed zoom=20 for every site regardless of size	**Adaptive zoom per site**, computed from the site's actual dimensions (`a_width`/`b_width`)
No cap on window size	Oversized linear "zanja" earthworks (up to 2,100m long) would otherwise demand a ~6,300m window at low resolution. Added `MAX_WINDOW_M = 800`, capping the crop to a detailed, recognizable segment rather than the full length at low detail
No HTTP status checking	Failed/blank tiles are now detected and logged, not silently stitched in
No throttling	Added per-tile and per-site delays to avoid triggering rate limits/blocking (1,443 sites × 64 tiles ≈ 92,000 requests)
Not resumable	Manifest-CSV-based resume: already-downloaded sites are skipped on rerun
Grayscale conversion by default	Dropped — no justification for discarding 2 of 3 color channels before training a classifier

Tile grid stays fixed at 8×8 (256px tiles), matching the prior team's `V2segmentationMaskGenerator.py` assumption — only the zoom (ground coverage) varies per site.

5. Positive-image download: execution and results

Run in Google Colab, cleaned CSV as input, output written directly to Google Drive (`geoglyph_v2/positives/`) so progress survives Colab disconnects.

- **Result: 1,436 of 1,443 sites downloaded successfully** (99.5%); 7 sites returned too many missing tiles.
- The run disconnected partway through (~1,250/1,443) and resumed cleanly from the Drive-backed manifest with no lost progress or duplicate work.
- **Manifest verification:** row counts in the manifest matched the actual file count on disk exactly (1,436 = 1,436) — no orphaned entries either way.
- **Visual spot-check (n=9, random sample):** most sites showed a clearly visible earthwork. One notable exception (`abuc1`, a 203m circle) showed only forest canopy — no visible geoglyph —

despite a technically clean download (0 missing tiles, correctly centered). This is consistent with the imagery-currency issue in Section 3: the site's defining clearing event (2021) may have since regrown.

Implication carried forward: a successful download does not guarantee a visually confirmed positive. A systematic verification step (e.g., mask-overlay verification, projecting known site geometry onto the image) is still needed before the full 1,436-image set can be trusted as ground truth — flagged as an open task.

6. Negative-image dataset: investigation (in progress)

Rather than generating a new negative set immediately, the existing 2,181 "non-geoglyph" images (from the original Zach/Colin dataset) were audited for reuse.

Contamination check, attempt 1 (filename code matching): failed — negative filenames follow the pattern `negative_<code>_<lat>_<lon>.png`, and a naive regex extracted the literal word "negative" instead of the embedded site code. Result: 0 flagged, which was correctly treated as unreliable rather than trusted at face value.

Contamination check, attempt 2 (coordinate proximity): parsed the actual lat/lon embedded in each negative's filename and computed the distance to the nearest official catalog site (BallTree, haversine).

Key finding: distances ranged from 67m to 484m across all 2,181 negatives (mean 265m), and most negatives' nearest official site matched the same code embedded in their own filename. **This reveals the prior team's negative-sampling strategy was not random background sampling — every negative was deliberately placed a small offset from a specific known geoglyph site**, i.e., a deliberate hard-negative / near-miss design.

This raises two open issues:

1. **38 negatives sit within 100m of a known site** — close enough, relative to some sites' actual size (a_width 100–200m+), that the real earthwork may bleed into the capture window. Visual inspection of these 38 confirmed **some genuinely contain a visible geoglyph** — i.e., real mislabeling, caught only by the coordinate-based check after the filename-based check missed it entirely.
2. **The existing negative set may have zero examples of plain, generic background** far from any known site — every one of the 2,181 is within 484m of a real geoglyph. If reused as-is, the model would never see an example of "ordinary forest with nothing nearby," which could limit generalization.

Status: manual removal of confirmed-contaminated images from the 38 flagged candidates is in progress (user reviewing directly in the Google Drive folder).

- **Ground-coverage matching:** positives use a variable window (150–800m, adaptive to site size). If negatives are captured at a different, less-varied window distribution, the model could learn to distinguish classes by zoom level/resolution rather than actual visual content (a shortcut-learning risk). Negative capture windows should be drawn from the same distribution as positives.
- **Negative diversity:** the reused hard-negative set (near-miss samples) should likely be supplemented with genuinely generic background negatives (plain forest, deforested-but-empty land) to avoid the gap noted in Section 6.
- **Pixel dimensions vs. ground coverage:** all images must be resized to one fixed pixel size at training time (architecture-dependent), which is independent of and does not resolve the ground-coverage-matching issue above.
- **Verification-before-labeling:** given the imagery-currency finding (Section 5) and the negative-mislabeling finding (Section 6), no downloaded image — positive or negative — should be treated as a confirmed label without some verification step. Manual review at full scale (thousands of images) was explicitly ruled out as infeasible; automated/coordinate-based checks plus targeted manual review of small flagged subsets is the adopted approach going forward.
- **Framing for the paper:** the model is intended to be recall-optimized with a human-in-the-loop triage framing (favor flagging possible sites for human review over missing real ones), which supports treating dataset construction itself as part of the methodological contribution, not just a preprocessing step.

8. Current dataset state (as of this log)

Item	Count	Status
Official catalog sites (raw)	1,477	—
Valid-coordinate sites	1,455	—
Cleaned sites (post dedup)	1,443	Final — `amazon_geoglyphs_cleaned.csv`
Positive images downloaded	1,436 / 1,443	Downloaded, spot-checked; full visual verification still pending

Item	Count	Status
Existing negative images	2,181	Under audit; 38 flagged near known sites, manual removal in progress

9. Next steps

1. Finish manual removal of confirmed-contaminated images from the 38 flagged negatives; re-verify folder count afterward.
2. Decide how to handle the negative-diversity gap (Section 6, item 2) — supplement with newly generated generic-background negatives.
3. Address ground-coverage/zoom matching between positives and negatives (Section 7) before finalizing the negative set.
4. Build a systematic visibility-verification step for the positive set (Section 5) rather than relying on random spot-checks alone.
5. Assemble a single manifest CSV (image path, label, site code, zoom/window, region group) covering both classes for reproducible, leakage-free model training and cross-validation.